

RPi Environmental Monitor – Project Description

Computer Systems and Networks

Student Name: Stephen Adamson

Student ID: W20119151

Project Description

This project involves the design and build of a small connected device. The Internet of Things (IoT) system designed will monitor the environmental conditions of its immediate surroundings such as temperature and humidity using a Raspberry Pi with SenseHat.

The system collects real time sensor on the Raspberry Pi at regular intervals. The data is processed to calculate average temperature and determine the comfort level of the environment (hot, cosy, or fine). The data is transmitted using an MQTT messaging protocol.

A second service receives the data, displays it in a terminal-based dashboard, and logs the information into a CSV file for later analysis.

Tools, Technologies and Equipment

The following list outlines the hardware, programming languages, and resources used in this project:

- Raspberry Pi 4
- Micro SD card
- SenseHAT sensor module
- Python 3
- MQTT (Mosquitto broker)
- Paho MQTT Python library
- JSON for data formatting and logging
- Git & GitHub for version control and documentation
- Linux (Raspberry Pi OS)

Project Repository

A Github repository has been created for this project (<https://github.com/StephenA112/IoT-Assignment>). This will contain the code and documentation for the project and also development commits.